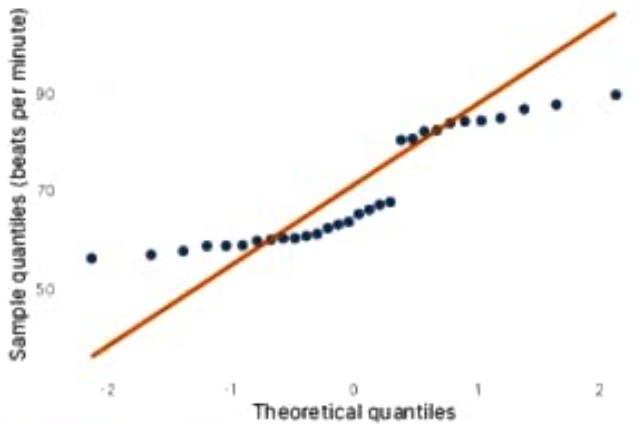
R.No: 25 Reg. No: 192473018 Name: N LAHARI

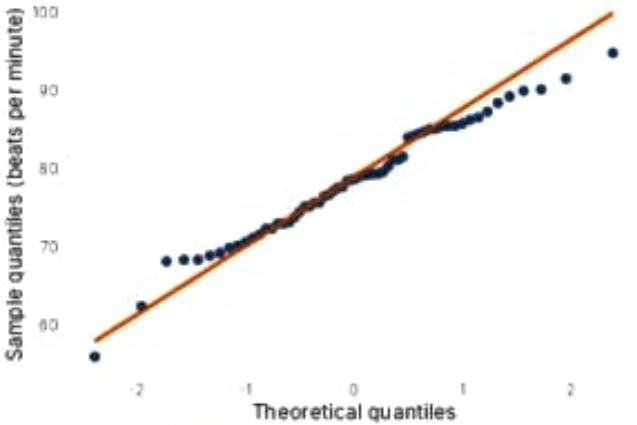
Each chart is a normal Q-Q plot made in R (stat_qq and stat_qq_line; red line = perfect normal fit). Part A (16 marks): for every chart decide whether the data are normal and name the shape from this list: Normal, Right-skewed, Left-skewed, Heavy-tailed, Light-tailed, Bimodal, Normal with outlier(s); then cite the evidence you see.

Chart 1: resting pulse (beats per minute), n = 30



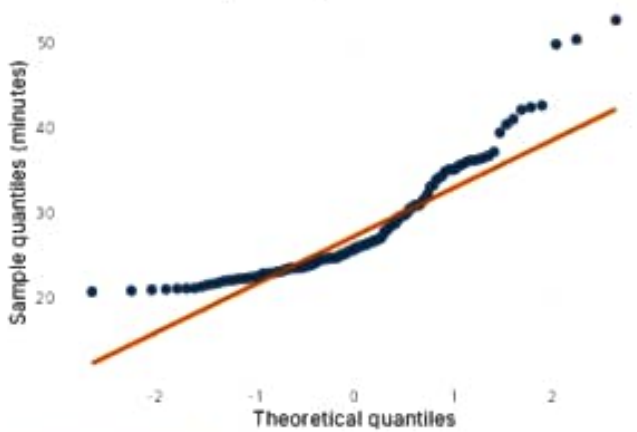
Normal? (Yes / No)	
Shape (from the list)	
Evidence in the plot	

Chart 2: resting pulse (beats per minute), n = 60



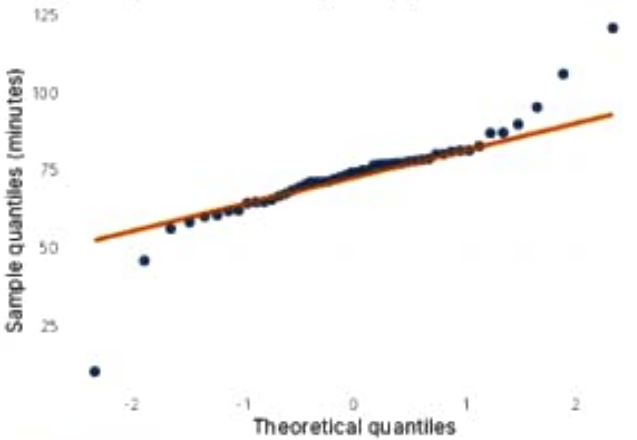
Normal? (Yes / No)	
Shape (from the list)	
Evidence in the plot	

Chart 3: commute time (minutes), n = 120



Normal? (Yes / No)	
Shape (from the list)	
Evidence in the plot	

Chart 4: lab experiment duration (minutes), n = 50



Normal? (Yes / No)	
Shape (from the list)	
Evidence in the plot	

Part B (4 marks)

B1. Chart 3 departs from normality. Name the departure and state the remedy you would try before using mean-and-SD methods (for example log transform, square, split the data, robust methods). [2]

B2. Fill the blanks to draw a Q-Q plot with its reference line for a data frame df with a column value: ggplot(df, aes(sample = value)) + _____ + _____ [2]

14/1/26 Activity-05

N. Lahari
192473018

Part A - Q-Q Plot Analysis

Chart 1: Resting pulse (n=30)

* Normal? No
* Shape: Bimodal
* Evidence in the plot: The points form two distinct groups with a clear jump between them and deviate from the reference line.

Chart 2: Resting pulse (n=60)

* Normal? Yes
* Shape: Normal
* Evidence in the plot: Most points lie close to the reference line, with only small deviations at the ends.

Chart 3: Commute time (n=120)

* Normal? No
* Shape: Right-skewed
* Evidence in the plot: the points curve upward above the reference line in the upper tail, indicating a long right tail.

Chart 4: Lab Experiment duration (n=50)

* Normal? No
* Shape: Heavy-tailed
* Evidence in the plot: The points at both ends deviate from the reference line, with extreme values in the tails, especially the upper tail.

Part B - Short Answers

B1. Chart 3 departs from normality. Name the departure & state the remedy.

* Departure: Right-skewed distribution.
* Remedy: Apply a log transformation to reduce right skewness before using mean-and-standard deviation methods.

B2. Fill in the blanks to draw a Q-Q plot.

```
ggplot(df, aes(sample = value)) +
    stat_qq() +
    stat_qq_line()
```

* stat_qq() - plots the theoretical & sample quantiles.
* stat_qq_line() - adds the reference line.